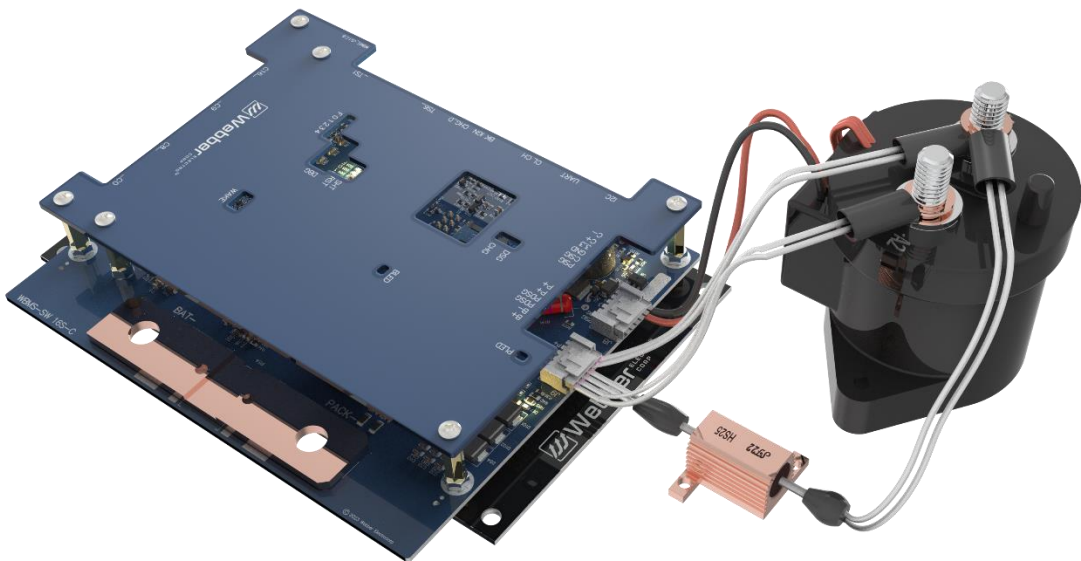
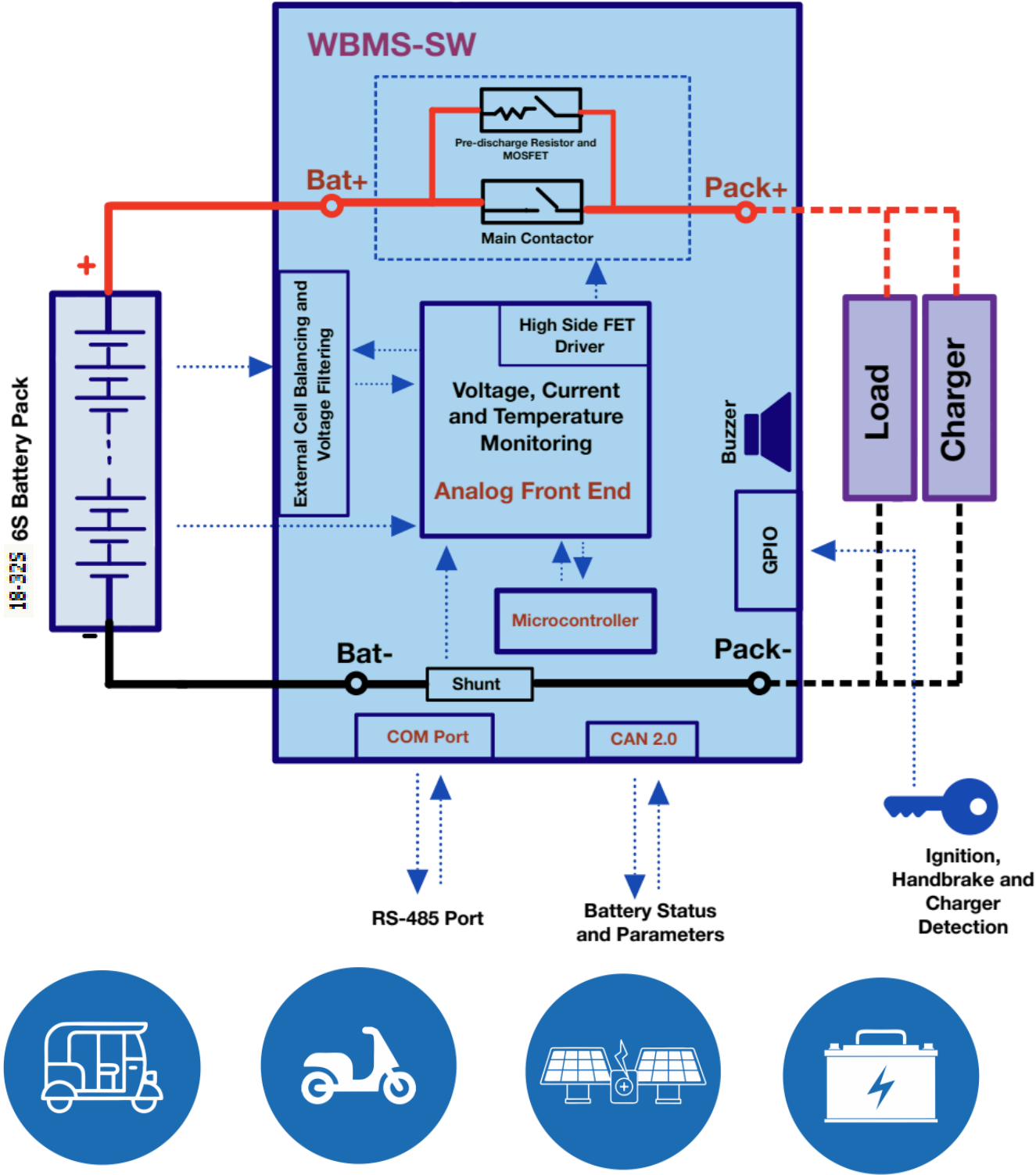


WBMS-SW-32S Contactor

**Smart BMS Manages Lithium Ion Batteries
with Precision and Speed**



Application Diagram



Salient Features

- Scalable up to **32 cells** & **75-140V** battery packs
- **AIS-156 PH-II Compliant**
- **Buzzer output** and **LED Indication** for **Thermal Runaway, Open Wire Fault**
- Multiple communication options: **Isolated CAN2.0** and **BLE**
- Onboard: **uSD Card** or **EEPROM**
- **High side Contactor Switching** provides safe and uninterrupted ground connection to external subsystems.
- **Hardware and Software interlocks** for Charge and Discharge Paths
- **On board power supply and switching or predischage control**. Does not need external 12V power supply and predisharge contactor
- **Predischage control** for preventing inrush currents to capacitive loads
- **Short Circuit and Overcurrent in discharge latches** ensure automatic locking of discharge path after repeated short circuit and overcurrent faults
- **High cell balancing current up to 380mA** minimizes cell to cell voltage and SOC variations and optimal charging of battery pack
- Individual Cell over and under voltage and temperature protection
- Pack level over current and short circuit protection
- **Deep Discharge Prevention (Auto BMS Shutdown)** when cell voltage is critically low
- Proprietary **CAN based field servicing tools** for **data monitoring, logging** and **firmware update**

Specifications		Remarks	
Number of Cells		Min: 18 Max: 32	
Cell Voltage Sensing Range		1.6 - 4.5V	
Peak Battery Terminal Voltage		140 V (32S)	
Sensing Accuracy		Cell Voltage: $\pm 5\text{mV}$ Current: $\pm 1\%$	
Dimension		165 mm X 102 mm X 28 mm (L X B X H)	
Nominal Current Rating	Charging and Discharging Current	Up to 150A	
Cell Balancing	Balancing Current per Cell	Up to $\sim 400\text{mA}$	
Pre-discharge control	Nominal Capacitive load	3-16mF in 0.5 seconds (Customisable)	
BMS Self Current Consumption	Active Mode, Shutdown Mode	Active Mode: $\sim 165\text{mA}$ (Contactor On) $\sim 85\text{mA}$ (Contactor OFF) Shutdown Mode: $\sim 1.3\mu\text{A}$	
Temperature Sensing			
	External	Twelve external channels for cell temperature	
	Range and Accuracy	Range: -10°C to 120°C Accuracy $\pm 1^{\circ}\text{C}$,	
Onboard Data Storage	uSD Card or EEPROM	uSD Card: every 0.5 second, logged over last 60 days	
Over Voltage Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF. Discharging is allowed if Charger is removed and Ignition is ON	
	Fault Release	Through Hysteresis, Automatic	
Under Voltage Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF Charging is allowed if Charger is connected and Ignition is OFF	
	Fault Release	Configurable Hysteresis	No load removal required. Charging current is allowed.
Over Current in Discharge Protection Level-1	Threshold, Protection Trigger Delay	Configurable to allow up to 400A discharge for up to 60 seconds	
	Fault Response	Contactor: OFF Charging and Discharging are allowed only after fault is cleared	
	Fault Release	Auto recovery after no discharge current for 90 secs	
Over Current in Discharge Protection Level-2	Threshold, Protection Trigger Delay	Configurable to instantaneously cut off at 900A discharge current in 6000 mSec	
	Fault Response	Contactor: OFF Charging and Discharging are allowed only after fault is cleared	
	Fault Release	Auto recovery after no discharge current for 90 secs	

Note: Specifications are indicative only. Final specifications and behaviour including continuous discharge/charge current, balancing current, protection thresholds, etc. may vary depending upon BMS version, BMS mounting scheme, PDU heat dissipation, programmed parameters and application logic.

Specifications			Remarks
Over Current in Discharge Latch Protection	Threshold	5 consecutive over discharge current faults	
	Fault Response	Contactor: OFF Charging and Discharging are allowed only after fault is cleared	
	Fault Release	Load Removal	
Over Current in Charge Protection Level-1	Threshold, Protection Trigger Delay	Configurable to allow up to 200A charge current for up to 60 seconds	
	Fault Response	Contactor: OFF Discharging is allowed if Charger is removed and Ignition is ON	
	Fault Release	Auto recovery after no charge current for 90 secs	
Over Current in Charge Protection Level-2	Threshold, Protection Trigger Delay	Configurable to instantaneously cut off at 250A charge current in 100 mSec	
	Fault Response	Contactor: OFF Discharging is allowed if Charger is removed and Ignition is ON	
	Fault Release	Auto recovery after no charge current for 90 secs	
Short Circuit in Discharge Protection	Protection Trigger Delay	~20 microSec	
	Fault Response	Contactor: OFF	
	Fault Release	Auto recovery after 5 seconds	
Short Circuit in Discharge Latch Protection	Threshold	2 consecutive short circuit faults	
	Fault Response	Contactor: OFF	
	Fault Release	Load Removal	
Over Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF Charging is allowed if Charger is connected and Ignition is OFF	
	Fault Release	Auto recovery below configurable temperature	
Under Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF Charging is allowed if Charger is connected and Ignition is OFF	
	Fault Release	Auto recovery above configurable temperature	
Over Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF Discharging is allowed if Charger is removed and Ignition is ON	
	Fault Release	Auto recovery below configurable temperature	
Under Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable	
	Fault Response	Contactor: OFF Discharging is allowed if Charger is removed and Ignition is ON	
	Fault Release	Auto recovery above configurable temperature	
Buzzer Output	Alert on Permanent Failure	Thermal Runaway, Contactor Failure, Cell Open Wire, Short Circuit Latch, Over Current in Discharge Latch, Insulation Failure	